Tier 4 | Multi-agent systems

Modules 15-18 | SDK 1.62 | 2026-10-29

Choose a pattern

One agent for a narrow task. Agents as tools for delegation. Graph for explicit dependencies. Swarm for bounded handoffs. A2A for independent remote agents.

Graph design

Name each node and its contract. Join independent evidence before computing totals. Set node execution, task and time limits. Return partial failure explicitly.

Delegation boundary

Give each specialist only its tools and required context. Validate specialist output against original evidence. Recalculate money in host code.

A2A trust

Read the agent card, authenticate the transport, check supported protocol, stream task status, validate artifacts and close owned processes. A card alone grants no trust.

Partner install

```
uv sync --group multiagent --group partner
uv run python checkpoints/04_trip_planning_crew/solution.py --mock --decision approve
```

Costs and concurrency

Count every specialist call and retry. Parallel work can reduce latency while increasing total cost. A shared budget must cover the entire request.

Test independently

Use held-out flights, budgets and malformed partner artifacts. Test timeouts, cross-user access and duplicate approval; do not grade a result only with its own producer.

Checkpoint

Build a bounded planning crew, show verified totals and a reviewed draft, then demonstrate an actual separate-process partner. No travel purchase is performed.